

Cálculo II - Lista 5 - Jeann

① Escreva as formas de decomposição em frações parciais das funções.

Não determine os valores numéricos dos coeficientes da decomposição

a) $P(x) = 1 + 6x = \text{Grau } 1$

(Função própria

$Q(x) = (4x-3) \cdot (2x+5) \rightarrow (8x^2 + 10x - 6x - 15) \rightarrow (8x^2 + 4x - 15) = \text{Grau } 2. \quad (1 < 2).$

$$\frac{A}{(4x-3)} + \frac{B}{(2x+5)}$$

b) $X / x^2 + x - 2$

$Px = X = \text{Grau } 1 \quad // \quad Qx = x^2 + x - 2 = \text{Grau } 2 \quad // \quad \text{Função Própria} = 1 < 2$

$(x^2 + x - 2) = (x+2) \cdot (x-1) \quad \text{logo}$

$$\frac{X}{x^2 + x - 2} = \frac{A}{(x+2)} + \frac{B}{(x-1)}$$

c) $4x / x^3 - x^2 - x + 1$

$Px = 4x = \text{Grau } 1 \quad // \quad Qx = x^3 - x^2 - x + 1 = \text{Grau } 3 \quad // \quad \text{Função Própria} = 1 < 3$

$x^3 - x^2 - x + 1 = (x-1)^2 \cdot (x+1)$

$$\frac{4x}{x^3 - x^2 - x + 1} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+1)}$$

d) $\frac{x^3 - x + 1}{x^2(x-1)^3} \quad Px = x^3 - x + 1 = \text{Grau } 3 \quad (\text{Função própria } 3 < 5.$

$Qx = (x^5 - 3x^4 + 3x^3 - x^2) \rightarrow \text{Grau } 5$

$$\frac{x^3 - x + 1}{x^2(x-1)^3} \rightarrow \frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-1)} + \frac{D}{(x-1)^2} + \frac{E}{(x-1)^3}$$

e) $1 - x + 2x^2 / x \cdot (x^2 + 1) \quad Px = 1 - x + 2x^2 = \text{Grau } 2 \quad // \quad Qx = x^3 + x = \text{Grau } 3 \quad // \quad 2 < 3.$

$$\frac{1 - x + 2x^2}{x \cdot (x^2 + 1)} \rightarrow \frac{A}{x} + \frac{Bx + C}{(x^2 + 1)}$$

f) $1 / (x^2 + 9)^3 \quad Px = 1 \quad \text{Grau } 0 \quad // \quad Q(x) = x^6 + 18x^4 + 81x^2 + 27 \quad // \quad 0 < 6.$

$$\frac{Ax + B}{(x^2 + 9)} + \frac{Cx + D}{(x^2 + 9)^2} + \frac{Ex + F}{(x^2 + 9)^3}$$

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2) Resolva as seguintes integrais usando as frações parciais

a) $\int \frac{x-9}{(x+5) \cdot (x-2)} dx$

$P_x = (x-9) = \text{Grau } 1 \parallel Q_x = (x+5) \cdot (x-2) = \text{Grau } 2 \parallel \text{Fracção Parcial } 1 < 2$

$$\frac{A}{(x+5)} + \frac{B}{(x-2)} \rightarrow \frac{A(x-2) + B(x+5)}{(x+5) \cdot (x-2)} \rightarrow \frac{Ax - 2A + Bx + 5B}{x^2 + 3x - 10}$$

$$\frac{(A+B)x + (-2A+5B)}{(x+5) \cdot (x-2)} = \frac{x-9}{(x+5) \cdot (x-2)}$$

$$\begin{cases} A+B=1 \\ -2A+5B=-9 \end{cases} \rightarrow \begin{cases} 2A+2B=2 \\ -2A+5B=-9 \end{cases} \rightarrow \begin{matrix} 7B=-7 & A=2 \\ B=-1 & B=-1 \end{matrix}$$

Vollstandig

$$\int \left(\frac{2}{x+5} - \frac{1}{x-2} \right) dx \rightarrow 2 \int \frac{dx}{x+5} - \int \frac{dx}{x-2} \rightarrow 2 \ln|x+5| - \ln|x-2| + C$$

A integral é: $2 \ln|x+5| - \ln|x-2| + C$

b) $\int_0^1 \frac{(x-4)}{x^2-5x+6} dx$

$P_x = (x-4) \rightarrow \text{Grau } 1 \parallel Q_x = (x^2-5x+6) \rightarrow \text{Grau } 2 \parallel \text{Fracção Parcial } 1 < 2$

$$\frac{A}{(x-3)} + \frac{B}{(x-2)} \rightarrow \frac{A(x-2) + B(x-3)}{(x^2-5x+6)} \rightarrow \frac{Ax - 2A + Bx - 3B}{(x^2-5x+6)} \rightarrow \frac{(A+B)x + (-2A-3B)}{(x^2-5x+6)}$$

$$\frac{(A+B)x + (-2A-3B)}{(x^2-5x+6)} = \frac{x-4}{(x^2-5x+6)} \rightarrow \begin{cases} A+B=1 \\ -2A-3B=-4 \end{cases} \rightarrow \begin{cases} 2A+2B=2 \\ -2A-3B=-4 \end{cases}$$

$-B=-2 \rightarrow B=2 \text{ e } A=-1$

Vollstandig

$$\int \left(\frac{-1}{x-3} + \frac{2}{x-2} \right) dx \rightarrow - \int \frac{dx}{x-3} + 2 \int \frac{dx}{x-2} \rightarrow -\ln|x-3| + 2 \ln|x-2| + C$$

Intervale 1.e 0

$-\ln 2 + 2 \ln 1 + \ln 3 - 2 \ln 2$

$-3 \ln 2 + 2 \ln 1 + \ln 3 \rightarrow -3 \ln 2 + \ln 3$

$$c) \int (x^2 + 2x - 1)/(x^3 - x) dx$$

$$P(x) = (x^2 + 2x - 1) = \text{Grau } 2 \parallel Q(x) = (x^3 - x) = \text{Grau } 3 \parallel \text{Função racional } 2 < 3$$

$$(x^3 - x) \rightarrow x \cdot (x^2 - 1) \rightarrow x \cdot (x+1) \cdot (x-1)$$

$$\frac{A}{x} + \frac{B}{(x+1)} + \frac{C}{(x-1)} \rightarrow \frac{A(x^2-1) + B(x^2+x) + C(x^2-x)}{(x^3-x)}$$

$$\frac{Ax^2 - A + Bx^2 + Bx + Cx^2 - Cx}{(x^3-x)} \rightarrow \frac{x^2(A+B+C) + x(B-C) - A}{(x^3-x)}$$

$$\begin{cases} A+B+C=1 \\ B-C=1 \\ -A=-1 \end{cases} \begin{cases} B=2+C \\ A=1 \\ C=-1 \end{cases}$$

Voltando

$$\int \left(\frac{1}{x} + \frac{1}{(x+1)} + \frac{-1}{(x-1)} \right) dx \rightarrow \int \frac{dx}{x} + \int \frac{dx}{(x+1)} - \int \frac{dx}{(x-1)}$$

$$\ln|x| + \ln|x+1| - \ln|x-1| + C$$

$$d) \int (x^2 - 5x + 16)/(2x+1) \cdot (x-2)^2 dx$$

$$P(x) = (x^2 - 5x + 16) = \text{Grau } 2 \parallel Q(x) = (2x+1) \cdot (x-2)^2 = \text{Grau } 3 \parallel \text{Função racional } 2 < 3$$

$$\frac{A}{(2x+1)} + \frac{B}{(x-2)} + \frac{C}{(x-2)^2} \rightarrow \frac{A(x-2)^2 + B(2x+1) \cdot (x-2) + C(2x+1)}{(2x+1)(x-2)^2}$$

$$(Ax^2 - A4x + A4) + (B2x^2 - B3x - 2B) + (C2x + C)$$

$$x^2(A+2B) + x(-4A-3B+2C) + (4A-2B+C)$$

$$\begin{cases} A+2B=1 \\ -4A-3B+2C=-5 \\ 4A-2B+C=16 \end{cases} \begin{cases} A+2B=1 \\ -4+8B-3B+2C=-5 \\ 4+8B-2B+C=16 \end{cases} \rightarrow \begin{cases} A+2B=1 \\ 5B+2C=-1 \\ -10B+C=12 \end{cases} \begin{cases} A+2B=1 \\ 5B+2C=-1 \\ 20B-2C=-24 \end{cases}$$

$$25B = -25 \parallel B = -1 \parallel A = 3 \parallel C = 2$$

$$\int \frac{3}{(2x+1)} - \frac{1}{(x-2)} + \frac{2}{(x-2)^2} \rightarrow 3 \int \frac{dx}{(2x+1)} - \int \frac{dx}{(x-2)} + 2 \int \frac{dx}{(x-2)^2}$$

$$3 \ln|2x+1| - \ln|x-2| + \int (x-2)^{-2} \rightarrow \frac{3 \ln|2x+1| - \ln|x-2| - 2}{(x-2)} + C$$

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$$e) \int (2x+3)/(x^3+6x^2+9x) dx$$

$$P(x) = (2x+3) dx \rightarrow \text{Grad 1} \parallel Q(x) = (x^3+6x^2+9x) dx \rightarrow \text{Grad 3} \parallel 1 < 3$$

$$x^3+6x^2+9x \rightarrow x(x+3)^2$$

$$\frac{A}{x} + \frac{B}{(x+3)} + \frac{C}{(x+3)^2} \rightarrow \frac{A(x+3)^2 + B(x)(x+3) + C(x)}{x(x+3)^2}$$

$$\frac{(Ax^2 + 6Ax + 9A) + (Bx^2 + 3Bx) + (Cx)}{x^2(A+B) + x(6A+3B+C) + 9A} \left\{ \int \left(\frac{113}{x} + \frac{113}{(x+3)} + \frac{1}{(x+3)^2} \right) dx \right.$$

$$\begin{aligned} A+B &= 0 & A &= 113 \\ 6A+3B+C &= 2 & B &= -113 \\ 9A &= 3 & C &= 1 \end{aligned} \left\{ \begin{aligned} &\frac{1}{3} \int \frac{dx}{x} - \frac{1}{3} \int \frac{dx}{(x+3)} + \int \frac{dx}{(x+3)^2} \end{aligned} \right.$$

$$\frac{1}{3} \ln|x| - \frac{1}{3} \ln|x+3| + \frac{1}{(x+3)} + C$$

$$F) \int 5/(x^2 \cdot (x-1)^2)$$

$$P(x) = 5 = \text{Grad 1} < 0 \parallel Q(x) = x^2 \cdot (x-1)^2 \rightarrow \text{Grad 4} \parallel 0 < 4$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-1)} + \frac{D}{(x-1)^2} \rightarrow \frac{A \cdot x \cdot (x-1)^2 + B(x-1)^2 + Cx^2 \cdot (x-1) + D \cdot x^2}{x^2(x-1)^2}$$

$$\frac{A(x^3-2x^2+x) + B(x^2-2x+1) + C(x^3-x^2) + D \cdot x^2}{x^3(A+C) + x^2(-2A+B-C+D) + x(A-2B) + B}$$

$$\begin{aligned} A+0B+C+0D &= 0 & A &= 10 \\ -2A+B-C+D &= 0 & B &= 5 \\ A-2B+0C+0D &= 0 & C &= -10 \\ B &= 5 & D &= 5 \end{aligned} \left\{ \int \left(\frac{10}{x} + \frac{5}{x^2} - \frac{10}{(x-1)} + \frac{5}{(x-1)^2} \right) dx \right.$$

$$\frac{10 \int \frac{dx}{x} + 5 \int \frac{dx}{x^2} - 10 \int \frac{dx}{(x-1)} + 5 \int \frac{dx}{(x-1)^2}}$$

$$10 \ln|x| - 5x^{-1} - 10 \ln|x-1| - 5(x-1)^{-1} + C$$

③ Simplifique as frações impróprias por meio da divisão entre polinômios e use frações parciais para resolver as integrais

a) $\int x / 5x + 6 \, dx$

Como não dá para simplificar mais essa fração, usarei o método

$\int x / 5x + 6 \, dx \quad u = 5x + 6, \, du/5 = dx$

$x = u/5$

Substituindo

$\int \frac{(u/5 + 6)/5 \cdot 1/5 \, du}{u} \rightarrow \frac{1}{25} \int \frac{(u/5 + 6)}{u} \, du \rightarrow \frac{1}{25} \int \frac{1}{u} \, du + \frac{6}{25} \int \frac{1}{u} \, du$

$\frac{1}{25} + \frac{6 \cdot \ln|u|}{25} + C \rightarrow \frac{5x+6}{25} + \frac{6 \cdot \ln|5x+6|}{25}$

b) $\int 3r^2 / 3r + 4 \, dr$

$\frac{3r^2}{3r+4} \quad \left| \begin{array}{l} 3r^2 \\ -3r^2 - 4r \\ \hline 4r \end{array} \right. \quad \left(\begin{array}{l} \text{Reescrevendo} \\ \int \frac{3r^2}{3r+4} \end{array} \right) \rightarrow \int \left(r - \frac{4r}{3r+4} \right) dr$

1º Integral $\int r \, dr$

$\int r \, dr \rightarrow r^2/2$

2º Integral

$\int 4r / 3r + 4 \, dr \rightarrow u = 3r + 4 \parallel du = 3 \, dr \parallel du/3 = dr \parallel r = (u - 4)/3$

Substituindo

$\int \frac{4 \cdot (u-4)/3}{u} \cdot \frac{1}{3} \, du \rightarrow \frac{4}{9} \int \left(1 - \frac{4}{u} \right) du \rightarrow \frac{4}{9} (u - 4 \ln|u|)$

$\frac{r^2}{2} - \frac{4}{9} (3r + 4 - 4 \ln|3r + 4|) + C$

$$c) \int (x^3 + 2x) / (2x - 1) dx$$

$$\begin{array}{r} x^3 + 2x \\ - x^3 + x^2/2 \\ \hline x^2/2 + 2x \end{array} \quad \begin{array}{r} (2x - 1) \\ - 12x^2 + 112x + 918 \end{array}$$

$$\begin{array}{r} x^2/2 + 2x \\ - x^2/2 + 112x \\ \hline 918 \end{array}$$

$$918/14$$

$$- 918/14 + 918$$

$$(918)$$

$$\int (x^3 + 2x) / (2x - 1)$$

$$\hookrightarrow \int x^2/2 + x/2 + 918 + 918/(2x - 1) dx$$

$$\downarrow$$

$$\frac{1}{2} \int x^2 dx + \frac{1}{2} \int x dx + \frac{9}{8} \int dx + \int \frac{918}{2x - 1} dx$$

$$\frac{x^3}{6} + \frac{x^2}{8} + \frac{9x}{8}$$

$$\int \frac{918}{2x - 1} dx \rightarrow \frac{9}{8} \int \frac{1}{2x - 1} dx \rightarrow \frac{1}{2} \frac{9}{8} \ln|2x - 1|$$

Juntando tudo.

$$\frac{x^3}{6} + \frac{x^2}{8} + \frac{9x}{8} + \frac{9}{16} \ln|2x - 1| + C$$

$$d) \int_0^1 (x^3 - 4x - 10) / (x^2 - x - 6) dx$$

$$\begin{array}{r} x^3 - 4x - 10 \\ - x^3 + x^2 + 6x \\ \hline x^2 + 2x - 10 \end{array} \quad \begin{array}{r} (x^2 - x - 6) \\ (x + 1) \end{array}$$

$$\begin{array}{r} x^2 + 2x - 10 \\ - x^2 + x + 6 \\ \hline 3x + 4 \end{array}$$

$$(3x + 4)$$

$$\int \frac{(3x + 4)}{(x^2 - x - 6)} \rightarrow (x^2 - x - 6) = (x - 3)(x + 2) \rightarrow \frac{A}{(x - 3)} + \frac{B}{(x + 2)}$$

$$\frac{A(x + 2) + B(x - 3)}{x^2 - x - 6} \rightarrow \frac{Ax + 2A + Bx - 3B}{x^2 - x - 6} \rightarrow (A + B)x + (2A - 3B)$$

$$\begin{cases} A + B = 3 \\ 2A - 3B = 4 \end{cases} \quad A = 1 \text{ e } B = 2$$

Voltando

$$\int \left(\frac{1}{(x - 3)} + \frac{2}{(x + 2)} \right) dx \rightarrow \int \frac{1}{x - 3} dx + \int \frac{2}{x + 2} dx \rightarrow \frac{1}{2} \int \frac{dx}{x - 3} + 2 \int \frac{dx}{(x + 2)} \rightarrow \frac{x^2}{2} + x + \ln|x - 3| + \ln|x + 2|$$

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Intervalo de la 0

$$(1/2 + 1 + \ln 2 + 2\ln 3 - \ln 3 - \ln 2)$$

$$\frac{3}{2} - \ln 2 + \ln 3$$

e) $\int (16x^3/4x^2-2x+1) dx$

$$\begin{array}{l} 16x^3 \\ - 16x^3 + 16x^2 - 2x \\ 16x^2 - 2x \\ - 16x^2 + 16x - 2 \\ (12x - 2) \end{array} \quad \begin{array}{l} | 4x^2 - 2x + 1 \\ (2x - 1) \\ (2x - 1) \end{array} \quad \begin{array}{l} 16x^3 / 4x^2 - 2x + 1 \\ \hookrightarrow 4x + 2 + (12x - 2) / (2x^2 - 2x + 1) \\ 2x^2 + 4x + \int (12x - 2) / (2x^2 - 2x + 1) dx \end{array}$$

$$\int (12x - 2) / (2x^2 - 2x + 1) dx \rightarrow (2x^2 - 2x + 1) \rightarrow (2x - 1)^2$$

$$\frac{A}{(2x-1)} + \frac{B}{(2x-1)^2} \rightarrow \frac{A(2x-1) + B}{(2x-1)^2} \rightarrow 2Ax - A + B$$

$$\begin{cases} \int 2A = 12 & A = 6 \\ -A + B = -2 & B = 2 \end{cases} \quad \left\{ \int \frac{6}{2x-1} + \frac{2}{(2x-1)^2} \rightarrow 6 \int \frac{1}{2x-1} + 2 \int \frac{1}{(2x-1)^2} \right.$$

$$6 \cdot \ln|2x-1| + 2 \cdot \frac{1}{2x-1} + 3 \ln|2x-1| + 2 \frac{1}{(2x-1)} + C$$

f) $\int (2y^4)/(y^3-y^2+y-1) dy$

$$\begin{array}{l} 2y^4 \\ - 2y^4 + 2y^3 - 2y^2 + 2y \\ 2y^3 - 2y^2 + 2y \\ - 2y^3 + 2y^2 - 2y + 2 \\ + 2 \end{array} \quad \begin{array}{l} | (y^3 - y^2 + y - 1) \\ (y^3 - y^2 + y - 1) \end{array} \quad \begin{array}{l} 2y^4 / (y^3 - y^2 + y - 1) \\ \hookrightarrow 2y + 2 + 2 / (y^3 - y^2 + y - 1) \\ y^2 + 2y + \int 2 / (y^3 - y^2 + y - 1) \end{array}$$

$$\int + 2 / (y^3 - y^2 + y - 1) \rightarrow (y-1) \cdot (y^2+1)$$

$$\frac{A}{(y-1)} + \frac{By+C}{(y^2+1)} \rightarrow \frac{A(y+1) + By+C(y-1)}{(y-1)(y^2+1)} \rightarrow Ay^2 + A + By^2 - By + Cy - C$$

$$\begin{cases} A + B = 0 & A = 1 \\ -B + C = 0 & B = -1 \\ A - C = 2 & C = -1 \end{cases} \quad \left\{ \frac{1}{(y-1)} + \frac{-y-1}{(y^2+1)} \right\} \quad \left\{ \ln|y-1| \right.$$

$$\int 1/(y-1) + \int (-y-1)/(y^2+1)$$

$$\rightarrow \ln|y-1| \rightarrow \int -y/(y^2+1) \text{ e } \int -1/(y^2+1)$$

$$-\int y/(y^2+1) \rightarrow u = (y^2+1), du = 2y dx \rightarrow du/2 = y dx$$

$$\frac{-1}{2} \int \frac{du}{u} \rightarrow \frac{-1}{2} \ln|u| \rightarrow \frac{-1}{2} \ln|y^2+1|$$

$$-\int \frac{1}{y^2+1} \rightarrow -\arctan(y) + C$$

Juntando tudo.

$$y^2 + 2y \frac{-1}{2} \ln|y^2+1| + \ln|y-1| - \arctan(y) + C$$

24) Calcule as integrais impróprias em intervalos ilimitados

$$a) \int_1^{\infty} x/(x^2+1)^2$$

$$\lim_{t \rightarrow \infty} \int_1^t x/(x^2+1)^2 dx \rightarrow u = (x^2+1), du = 2x dx$$

$$\rightarrow \frac{1}{2} \int \frac{du}{u^2} \rightarrow \frac{1}{2} \ln|u| \rightarrow \frac{1}{2} \ln|x^2+1| \dots$$

$$\lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln(x^2+1) \right]_1^t = \lim_{t \rightarrow \infty} \left(\frac{1}{2} \ln(t^2+1) - \frac{1}{2} \ln 2 \right)$$

$$\lim_{t \rightarrow \infty} \left(\frac{1}{2} \ln \infty - \frac{1}{2} \ln 2 \right) \rightarrow \lim_{t \rightarrow \infty} \infty - \frac{1}{2} \ln 2 \rightarrow \infty$$

A integral diverge.

$$b) \int_2^{\infty} 1/(x-2)^{3/2} dx$$

$$\lim_{t \rightarrow \infty} \int_2^t 1/(x-2)^{3/2} dx \rightarrow u = (x-2), du = dx \rightarrow \int du/u^{3/2} \rightarrow \int u^{-3/2} du$$

$$\frac{u^{-1/2}}{-1/2} \rightarrow \frac{-2}{u^{1/2}} \rightarrow \frac{-2}{(x-2)^{1/2}} \text{ Aplicar limites}$$

$$\frac{-2}{(t-2)^{1/2}} + 2 \rightarrow \frac{-2}{\infty} + 2 \rightarrow 2$$

A integral converge e tem valor 2.

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$$c) \int_0^{\infty} x^2 / \sqrt{1+x^3} dx$$

$$\lim_{t \rightarrow \infty} \int_0^t x^2 / \sqrt{1+x^3} dx \rightarrow u = (1+x^3) \rightarrow du = 3x^2 \rightarrow du/3 = x^2$$

$$\frac{1}{3} \int_0^t u^{-1/2} du \rightarrow \frac{1}{3} \cdot \frac{u}{-1/2} \rightarrow \frac{1}{3} \cdot 2u \rightarrow \frac{2}{3} (1+x^3)^{1/2}$$

$$\lim_{t \rightarrow \infty} t = 0$$

$$\frac{2}{3} ((1+t^3)^{1/2} - (1)) \rightarrow \frac{2}{3} (1+\infty)^{1/2} - 1 \rightarrow \infty$$

Esta integral diverge.

$$d) \int_{-\infty}^0 x \cdot e^{-x^2} dx$$

$$\lim_{a \rightarrow -\infty} \int_a^0 x e^{-x^2} dx + \lim_{b \rightarrow \infty} \int_0^b x e^{-x^2} dx \quad u = -x^2, du/2 = x dx$$

$$\frac{1}{2} \int_a^0 e^u \cdot du \rightarrow -\frac{1}{2} e^{-x^2} + C \rightarrow -\frac{1}{2} e^{-a^2} - \frac{1}{2} e^0 \rightarrow -\frac{1}{2}$$

$$\frac{1}{2} \int_0^b e^u \cdot du \rightarrow \frac{1}{2} e^{-x^2} + C \rightarrow \frac{1}{2} e^{-0^2} - \frac{1}{2} e^{-b^2} \rightarrow \frac{1}{2}$$

$$\frac{1}{2} - \frac{1}{2} = 0$$

Portanto ela converge.

$$e) \int_{-\infty}^0 r e^r dr$$

$$\lim_{t \rightarrow -\infty} \int_t^0 r e^r dr \rightarrow u = r, du = dr, dv = e^r dx, v = e^r$$

$$\int u \cdot dv = r \cdot e^r - \int e^r \cdot dr \rightarrow r \cdot e^r - e^r \rightarrow e^r (r-1) + C$$

Aplicando intervalos

$$e^0 \cdot (-\infty - 1) - e^0 \cdot (-1) \rightarrow (-1) - 0 = -1$$

A integral converge.

$$f) \int_0^{\infty} 2e^{-x} \cdot \sin x dx$$

$$\lim_{t \rightarrow \infty} \int_0^t 2e^{-x} \cdot \sin x dx \rightarrow 2 \int_0^t e^{-x} \cdot \sin x dx, u = \sin x, du = \cos x dx, dv = e^{-x} dx, v = -e^{-x}$$

$$\int u \cdot dv = \sin x \cdot (-e^{-x}) + \int \cos x \cdot e^{-x} dx \rightarrow u = \cos x, du = -\sin x dx, dv = e^{-x}, v = -e^{-x}$$

$$-\cos x \cdot e^{-x} - \int \sin x \cdot e^{-x} dx$$

$$\text{Voltando} \rightarrow \int e^{-x} \sin x dx = -\sin x \cdot e^{-x} - \cos x \cdot e^{-x} - \int e^{-x} \sin x dx$$

$$2 \int e^{-x} \sin x dx = -\sin x \cdot e^{-x} - \cos x \cdot e^{-x} \rightarrow \frac{-\sin x \cdot e^{-x} - \cos x \cdot e^{-x}}{2}$$

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Aplicando os intervalos

$$-\frac{\text{den } e^x - \cos \cdot e^x}{2} \Big|_0^\infty \rightarrow -\frac{\text{den } e^\infty - \cos \cdot e^\infty}{2} + \frac{\text{den } e^0 + \cos \cdot e^0}{2}$$

$$-\frac{1}{2} (\text{den } 1 + \cos 1) \rightarrow -\frac{1}{2}$$

Portanto converge.

g) $\int_{-\infty}^{\infty} 4 dx / x^2 + 16$

$$\lim_{a \rightarrow -\infty} \int_a^0 4 dx / x^2 + 16 + \lim_{b \rightarrow \infty} \int_0^b 4 dx / x^2 + 16$$

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$$\int_a^0 4 dx / x^2 + 16 \rightarrow \int_a^0 \frac{dx}{(x^2 + 16)} \rightarrow \frac{4}{4} \cdot \frac{1}{4} \arctan(x/4) + C$$

$$\arctan(x/4) \rightarrow \text{Aplicando limites} \rightarrow \arctan(0) - \arctan(\infty) \rightarrow +\pi/2$$

$$\arctan(x/4) \rightarrow \text{Aplicando limites} \rightarrow \arctan(\infty) - \arctan(0) \rightarrow +\pi/2$$

$$\frac{\pi}{2} + \frac{\pi}{2} \rightarrow \pi$$

h) $\int_1^{\infty} dx / x \sqrt{x^2 - 1}$

$$\lim_{t \rightarrow \infty} \int_1^t dx / x \sqrt{x^2 - 1} \rightarrow u = \sec \theta, du = \sec \theta \cdot \tan \theta d\theta, \sqrt{x^2 - 1} = \tan \theta$$

$$\int_0^t \frac{\sec \theta \cdot \tan \theta d\theta}{\sec \theta \cdot \tan \theta} \rightarrow \int_1^t d\theta \rightarrow \theta + C \rightarrow \text{arcsec } x + C$$

$$\text{arcsec } \infty - \text{arcsec } 1 \rightarrow \frac{\pi}{2} - 0 \rightarrow \frac{\pi}{2}$$

5)

Calcule as integrais impróprias.

a) $\int_2^3 \frac{1}{\sqrt{3-x}}$

$$\frac{1}{\sqrt{3-x}} \rightarrow u = (3-x), du = -1dx \rightarrow \int \frac{du}{u^{1/2}} \rightarrow \int u^{-1/2} \rightarrow \frac{2u^{1/2}}{-2(3-x)^{1/2}}$$

Aplicar limites

$$\lim_{b \rightarrow 3^-} [-2 \cdot (3-x)^{1/2}]_2^b \rightarrow -2 \cdot (3-b)^{1/2} + 2 \cdot (3-2)^{1/2} \rightarrow -2 \cdot (3-b)^{1/2} + 2$$

$0 + 2 \rightarrow$ A integral converge para 2.

b) ~~$\int_0^1 \frac{dx}{(1-x)}$~~ $\int_0^1 \frac{dx}{(1-x)}$

$$\int_0^1 \frac{dx}{(1-x)} \rightarrow -\ln|1-x|$$

Em 1 a função é indefinida, portanto separe a integral

$$\lim_{a \rightarrow 1^-} [-\ln|1-x|]_0^a \text{ e } \lim_{b \rightarrow 1^+} [-\ln|1-x|]_b^1$$

$$(-\ln|1-1| + \ln|1-0|) \rightarrow -\ln 0 + \ln 1 = +\infty$$

$$(-\ln|1-2| + \ln|1-1|) \rightarrow -\ln 1 + \ln 0 \rightarrow -\infty$$

A integral diverge

c) $\int_{-1}^1 \frac{dx}{x^{2/3}}$

$$\int_{-1}^1 \frac{dx}{x^{2/3}} \rightarrow \int_{-1}^1 x^{-2/3} dx \rightarrow 3x^{1/3} + C$$

A integral é indefinida no 0, portanto tenha que separar entre 2 limites

$$\lim_{b \rightarrow 0^+} [3x^{1/3}]_{-1}^b \rightarrow (3b^{1/3} - 3 \cdot (-1)^{1/3}) \rightarrow 3 \cdot 0 - (-3) \rightarrow 3$$

$$\lim_{a \rightarrow 0^+} [3x^{1/3}]_a^1 \rightarrow 3 \cdot 1^{1/3} - (3 \cdot a^{1/3}) \rightarrow 3 - 0 \rightarrow 3$$

Portanto ela converge em 6.

d) $\int_0^1 x \ln x \, dx$
 $\int_0^1 x \ln x \, dx \rightarrow u = \ln x, du = 1/x \, dx, dv = x \, dx, v = x^2/2$

$$\ln x \cdot \frac{x^2}{2} - \int \frac{x^{-1}}{2} \cdot x^2 \rightarrow \ln x \cdot \frac{x^2}{2} - \frac{1}{2} \int x \, dx$$

$$\left(\ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} \right) \left\{ \text{Em } 0, \text{ a integral é indefinida} \right.$$

$$\lim_{b \rightarrow 0^+} \left[\ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} \right]_b^1 \rightarrow \ln 1 \cdot \frac{1}{2} - \frac{1}{4} - \left(\ln b \cdot \frac{b^2}{2} + \frac{b^2}{4} \right) \rightarrow 0$$

Portanto a integral converge para $-1/4$.

e) $\int_1^b \frac{dx}{\sqrt{|x|}}$

$$\int_1^b (|x|)^{-1/2} dx \rightarrow 2|x|^{1/2}$$

A integral é indefinida em 0, portanto temos que separar em dois limites.

$$\lim_{b \rightarrow 0^+} [2|x|^{1/2}]_1^b \rightarrow 2\sqrt{b} + 2 \rightarrow +2$$

$$\lim_{a \rightarrow 0^+} [2|x|^{1/2}]_a^4 \rightarrow 2\sqrt{4} - 2\sqrt{a} \rightarrow 4 - 0 \rightarrow 4$$

$$4 + (+2) \rightarrow 6$$

f) $\int_0^1 (x+1)/\sqrt{x^2+2x} \, dx$

$$u = (x^2+2x), du = 2x+2 \rightarrow du/2 = (x+1) \, dx$$

$$\frac{1}{2} \int \frac{du}{u^{1/2}} \rightarrow \frac{1}{2} \int u^{-1/2} du \rightarrow \frac{1}{2} \cdot 2 u^{1/2} \rightarrow (x^2+2x)^{1/2}$$

$$\lim_{b \rightarrow 0^+} (x^2+2x)^{1/2} \Big|_b^1 \rightarrow (1^2+2)^{1/2} - (b^2+2b)^{1/2} \rightarrow \sqrt{3} - 0 \rightarrow \sqrt{3}$$

A integral converge para $\sqrt{3}$.

$$g) \int_0^2 (s+1) / \sqrt{4-s^2} ds$$

$$\int_0^2 (s) / \sqrt{4-s^2} + \int_0^2 1 / \sqrt{4-s^2}$$

$$1^\circ + \text{Integral} \rightarrow u = 4-s^2, du = -2s ds \rightarrow -du/2 = s ds$$

$$\frac{1}{2} \int_0^2 du / u^{1/2} \rightarrow -1/2 \int_0^2 u^{-1/2} \rightarrow -1/2 \cdot 2u \rightarrow (-4+s^2)$$

$$\int_0^2 1 / \sqrt{4-s^2} ds \rightarrow \arcsen(s/2) \dots$$

Aplicando limites

$$\lim_{b \rightarrow 2^-} [-\sqrt{4-s^2}]_0^b \rightarrow -\sqrt{4-b^2} + \sqrt{4-0} \rightarrow 0+2 \rightarrow 2$$

$$\lim_{a \rightarrow 2^-} [\arcsen(s/2)]_0^a \rightarrow \arcsen(a/2) - \arcsen 0 \rightarrow \frac{\pi}{2} - 0 \rightarrow \left(\frac{\pi}{2}\right)$$

A integral converge em $2 + \frac{\pi}{2} \rightarrow$

$$h) \int_0^\infty dx / ((1+x) \cdot \sqrt{x}) \quad u = \sqrt{x}, du = 1/(2\sqrt{x}) \rightarrow 2du = dx / \sqrt{x}$$

$$2 \int_0^\infty \frac{u dx}{(1+x) \cdot u} \rightarrow 2 \int_0^\infty \frac{du}{(1+u^2)}$$

Resolver por substituição trigonométrica $\int_0^\infty du / (1+u^2)$

$$\rightarrow u = \tan \theta, du = \sec^2 \theta d\theta$$

$$\int_0^\infty \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)} \rightarrow \int_0^\infty \frac{\sec^2 \theta d\theta}{\sec^2 \theta} \rightarrow \int_0^\infty d\theta \rightarrow 9$$

$$\lim_{t \rightarrow \infty} 2 \cdot [\theta]_0^\infty \rightarrow 2 \cdot (\arctan \sqrt{x} - \arctan \sqrt{0})$$

$$\rightarrow 2 \cdot (\pi/2 - 0)$$

$$\rightarrow \pi$$

6. Considere a função $f(x) = (2x^3)/\sqrt{x^4+1}$, onde x é uma variável real.

a) Mostre que $\int_0^\infty f(x)dx$ diverge e, portanto, $\int_{-\infty}^\infty f(x)dx$ diverge.

$$f(x) = (2x^3)/\sqrt{x^4+1} \quad u = (x^4+1), \quad du = 4x^3 \rightarrow du/4 = x^3 dx$$

$$\int_0^\infty \frac{2 \cdot du \cdot 1/4}{(u)^{1/2}} \rightarrow \frac{1}{2} \int u^{-1/2} du \rightarrow \frac{1}{2} \cdot 2u^{1/2} \rightarrow u^{1/2} \dots$$

$$(x^4+1)^{1/2}$$

Aplicando o limite

$$\lim_{t \rightarrow \infty} \rightarrow [(x^4+1)^{1/2}]_0^t \rightarrow (t^4+1)^{1/2} - (0+1)^{1/2}$$

$$\rightarrow \infty - 1$$

$$\rightarrow \infty$$

A integral diverge.

b) Mostre que $\lim_{b \rightarrow \infty} \int_{-b}^b f(x)dx = 0$

Como a integral anterior é a mesma irei reutilizar

$$\lim_{b \rightarrow \infty} [(x^4+1)^{1/2}]_{-b}^b \rightarrow \lim_{b \rightarrow \infty} ((b^4+1)^{1/2} - ((-b)^4+1)^{1/2})$$

$$\rightarrow \infty - \infty$$

$$\rightarrow 0$$

Portanto $\lim_{b \rightarrow \infty} \int_{-b}^b f(x)dx = 0$.

7) Considere $a > 0$ e $n \in \mathbb{N}$

a) Mostre que $\int_0^\infty e^{-ax} dx = 1/a$. $u = -ax, du = -a dx, -du/a = dx$

$$\int e^u \cdot du \cdot \frac{1}{a} \rightarrow \frac{1}{a} \int e^u \rightarrow \frac{1}{a} \cdot e^u \rightarrow \frac{1}{a} \cdot e^{-ax}$$

Aplicando o limite

$$\lim_{t \rightarrow \infty} [-1/a \cdot e^{a \cdot x}]_0^\infty \rightarrow \frac{-1}{a} \cdot e^{a \cdot t} - (\frac{-1}{a} \cdot e^{a \cdot 0})$$

$$\frac{-1}{a} \cdot e^{a \cdot \infty} - (\frac{-1}{a} \cdot 1) \rightarrow 0 + \frac{1}{a} \rightarrow \frac{1}{a}$$

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b) Use integração por partes e o item a) para mostrar que $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$.

$$I_n = \int_0^\infty x^n e^{-ax} dx \quad u = x^n, dv = e^{-ax} dx, du = n \cdot x^{n-1} dx \quad e \quad v = -\frac{1}{a} e^{-ax}$$

$$I_n = \left[-x^n/a \cdot e^{-ax} \right]_0^\infty + n/a \int_0^\infty x^{n-1} e^{-ax} dx$$

$$\lim_{x \rightarrow 0} x^n/a \cdot e^{-ax} = 0$$

$$\lim_{x \rightarrow \infty} x^n/a \cdot e^{-ax} = 0$$

$$I_n = 0 + \frac{n}{a} I_{n-1}$$

$$I_n = \frac{n}{a} I_{n-1}$$

$$I_0 = \int_0^\infty e^{-ax} dx = 1/a$$

$$I_n = \frac{n}{a} \cdot \frac{(n-1)!}{a^{n-1}} = \frac{n!}{a^{n+1}}$$

8) Use o teste da comparação para mostrar a divergência ou convergência da integral

$$a) \int_1^\infty dx/x^3 + 1$$

$$\text{Comparação } dx/(x^3+1) \text{ e } dx/x^3$$

$$dx/(x^3+1) < dx/x^3 \text{ para } x \geq 1$$

$$\int_1^\infty \frac{dx}{x^3} \rightarrow \frac{x^{-2}}{-2} \rightarrow \left[-\frac{1}{2} x^{-2} \right]_1^\infty = 1/2$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{2} x^{-2} \right]_1^t \rightarrow -\frac{1}{2} - \left(-\frac{1}{2} \right) = 0$$

Integral converge para 1/2

Logo se $g(x)$ converge, $f(x)$ também converge.

$$b) \int_1^\infty dx/\sqrt{x} - 1$$

$$\text{Comparação } dx/\sqrt{x} - 1 \text{ e } dx/\sqrt{x}$$

$$dx/\sqrt{x} - 1 \geq dx/\sqrt{x} \text{ para } x \geq 1$$

$$\int_1^\infty dx/\sqrt{x} \rightarrow 2\sqrt{x}$$

$$\lim_{t \rightarrow \infty} [2\sqrt{x}]_1^t \rightarrow 2\sqrt{t} - 2\sqrt{1} = \infty - 2 = \infty$$

Integral diverge

Logo a integral $\int_1^\infty dx/\sqrt{x} - 1$ diverge.

$$c) \int_6^\infty dx/\sqrt{x^2+1}$$

$$\text{Comparação } dx/\sqrt{x^2+1} \text{ e } dx/\sqrt{x^2}$$

$$dx/\sqrt{x^2+1} > dx/x$$

$$\int \frac{dx}{x} \rightarrow \ln|x|$$

$$\lim_{t \rightarrow \infty} [\ln|x|]_6^t \rightarrow \ln|t| - \ln|6| \rightarrow \ln \infty - \ln 6 \rightarrow \infty$$

Logo $f(x)$ diverge.



d) $\int_2^{\infty} 1/\ln x \, dx$

Comparação $1/\ln x \, dx$ e $1/x^b \, dx$ com $b = 2 \dots$

$$\lim_{t \rightarrow \infty} [x^{-b+1}/(-b+1)]_2^t = \lim_{t \rightarrow \infty} \frac{t^{-1} - 2^{-1}}{-1 - -1} = \infty$$

$1/\ln x \, dx > 1/x^b \, dx$

$\int 1/x^b \, dx \rightarrow \int x^{-b} \, dx \rightarrow \frac{x^{-b+1}}{-b+1}$

A integral Diverge.

e) $\int_1^{\infty} e^x/x \, dx$

Comparação $e^x/x \, dx$ com $e^x \, dx$

$$\lim_{t \rightarrow \infty} |e^x|_1^t \rightarrow e^t - e^1 \rightarrow \infty$$

$\int e^x \rightarrow e^x$

Portanto como $g(x)$ diverge.
 $f(x)$ também vai divergir.

f) $\int_0^{\pi} \sin x / \sqrt[3]{x} \, dx$

$\int_0^{\pi} \sin x \, dx$ Comparação: $0 < \sin x < 1$ ($0 < x < \pi$)

$$0 < \frac{\sin x}{\sqrt[3]{x}} < \frac{1}{\sqrt[3]{x}}$$

$$\int_0^{\pi} \frac{\sin x}{\sqrt[3]{x}} \, dx < \int_0^{\pi} \frac{1}{\sqrt[3]{x}} \, dx = \frac{4}{3} \pi^{3/4}$$

$$\lim_{t \rightarrow 0^+} \frac{1}{\sqrt[3]{x}} \Big|_0^{\pi} = \lim_{t \rightarrow 0^+} \frac{4}{3} (\pi^{3/4} - t^{3/4})$$

g) $\int_0^{\infty} (2x^3 + 3x + 1)/(3x^4 + 5x^3 + 2x + 2) \, dx$

Comparar $(2x^3 + 3x + 1)/(3x^4 + 5x^3 + 2x + 2)$

$$\int 2x^3/3x^4 \rightarrow 2/3 \int 1/x^3 \rightarrow 2/3 \int x^{-3} \rightarrow -2/3 x^{-2} = -2/3 x^{-1}$$

Aplicando Limite

$$\lim_{t \rightarrow \infty} [-2/3 x^{-1}]_0^t \rightarrow \frac{-2}{3} \cdot \frac{1}{t} - 0 = 0$$

Logo a integral converge.

9) Calcule as integrais impróprias usando frações parciais.

a) $\int_{-1}^{\infty} d\theta / \theta^2 + 5\theta + 6$.

$\int dx / x^2 + 5x + 6 \rightarrow (x^2 + 5x + 6) \rightarrow (x+2) \cdot (x+3)$

$$\frac{A}{(x+2)} + \frac{B}{(x+3)} \rightarrow \frac{A(x+3) + B(x+2)}{(x+2) \cdot (x+3)} \rightarrow \frac{Ax + 3A + Bx + 2B}{(x+2) \cdot (x+3)}$$

$$\begin{cases} A + B = 0 & A = -B \\ 3A + 2B = 1 & -3B + 2B = 1 \end{cases} \rightarrow \begin{cases} -B = 1 \rightarrow B = -1 \\ A = 1 \end{cases}$$

$\int \left(\frac{1}{(x+2)} - \frac{1}{(x+3)} \right) dx \rightarrow \ln|x+2| - \ln|x+3|$

Aplicando os Limites

$$\begin{aligned} \lim_{t \rightarrow \infty} [\ln|x+2| - \ln|x+3|] \Big|_{-1}^t &\rightarrow \ln|t+2| - \ln|t+3| - \ln|1| + \ln|2| \\ &\rightarrow \infty - \infty - 0 + \ln 2 \\ &\rightarrow -\ln 2 \end{aligned}$$

O resultado da integral é $\ln 2$.

b) $\int_0^{\infty} dx / (x+1) \cdot (x^2+1)$

$\int dx / (x+1) \cdot (x^2+1) \rightarrow \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+1)} \rightarrow \frac{A \cdot (x^2+1) + Bx+C \cdot (x+1)}{(x+1) \cdot (x^2+1)} \rightarrow \frac{Ax^2 + A + Bx^2 + Bx + Cx + C}{(x+1) \cdot (x^2+1)}$

$$\begin{cases} A + B = 0 & A + B = 0 & 2C = 1 \rightarrow C = 1/2 \\ B + C = 0 & \rightarrow B + C = 0 \rightarrow B = -1/2 \\ A + C = 1 & -B + C = 1 & A = 1/2 \end{cases}$$

$\int \left(\frac{1/2}{(x+1)} + \frac{(-1/2)x + 1/2}{(x^2+1)} \right) \rightarrow \frac{1}{2} \int \frac{1}{(x+1)} - \frac{1}{2} \int \frac{x}{(x^2+1)} + \frac{1}{2} \int \frac{1}{(x^2+1)}$

1º: $(\ln|x-1|)/2$ //

2º: $1/2 \int x / (x^2+1) \rightarrow u = (x^2+1), du/2 = x dx$

$-1/4 \int du/u \rightarrow -1/4 \ln|u| \rightarrow -1/4 \ln|x^2+1|$

3º: $1/2 \int 1/(x^2+1) \rightarrow 1/2 \arctan x$

Aplicando Limites $\rightarrow \ln|(x-1)|/2 - 1/4 \ln|x^2+1| + \arctan x / 2$. $\lim_{t \rightarrow \infty}$

$(\ln|(t-1)|/2 - 1/4 \ln|t^2+1| + \arctan t / 2) - (\ln|(-1)|/2 + 1/4 \ln|1| + \arctan 0 / 2)$

$\infty - \infty + \frac{\pi}{4} - 0 + 0 + 0 \rightarrow \frac{\pi}{4}$

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